Soft Mobility Theory

Christophe Eloy
Aix-Marseille Univ, CNRS, Centrale Med, IRPHE, Marseille, France[*] and
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, UK
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Elastohydrodynamics is the interaction of elastic bodies with viscous flow. It governs phenomena from molecular translocation to microorganism locomotion. While the continuum formulation is well-established, analytical solutions exist only in asymptotic limits, and numerical approaches are typically problem-specific. We present a general theoretical framework for elastohydrodynamic systems that admit finite-dimensional parameterization through generalized coordinates describing both rigid-body motion and deformation. Applying d'Alembert's principle, we derive a soft mobility equation that extends classical rigid-body mobility theory to configuration-dependent systems. The mobility, advection, and flow-gradient coupling tensors all depend on the instantaneous configuration, yielding a nonlinear dynamical system amenable to efficient computation and gradient-based optimization. We demonstrate the framework through assemblies of hydrodynamically-interacting spheres connected by elastic springs, where the Rotne-Prager-Yamakawa tensor provides the configuration-dependent mobility. Three canonical problems reveal new physics: elastic corrections to Jeffery orbits exhibit memory effects and flow sensing through deformation; magnetic actuation produces efficient swimming via resonant deformation; and passive particles steer autonomously through preferential sampling of flow structures. This soft mobility formulation provides a unified approach to diverse elastohydrodynamic phenomena, from biological microswimmers to synthetic soft microrobots.

I. INTRODUCTION

Microorganisms rely on their deformability to interact with the surrounding fluid [1, 2]. This deformation can be active when microswimmers propel themselves through their slender flagella and cilia [2–6]. It can also be passive, like red blood cells undergoing large shape changes as they flow through capillaries [7, 8], or diatoms using flexible structures such as mucilage stalks and fine spines to adapt their sinking velocity [9–11]. Across these systems, deformability is not incidental but functional, often conferring sensing, locomotion, or feeding advantages that a rigid body could not achieve.

In artificial systems, flexible fibers and filaments have been shown to buckle and tumble in quiescent, shear, and turbulent flows [12–15], and a growing class of soft microrobots is being engineered for biomedical applications, such as in-situ sensing, drug delivery, and microsurgery [16–22]. Even microplastics deform and disperse in the ocean in ways that depend on their shape and flexibility [23].

These problems share a common physical structure: a deformable body whose shape evolves under the combined action of active, elastic, viscous, and body forces. This broad class of fluid–structure interactions is conventionally referred to as *elastohydrodynamics*. We adopt here the more specific term *soft mobility*, by which we mean the transport and deformation of a small deformable body immersed in a background flow, in direct analogy with the classical rigid-body mobility theory [24–26].

Soft mobility encompasses both the *forward* problem of predicting how a given soft body moves and deforms in a prescribed flow, and the *inverse* problem of designing soft bodies to perform a target function: efficient locomotion, flow sensing, clustering, or autonomous navigation in complex environments. The latter is increasingly central to soft robotics [17, 27] and embodied intelligence [28], and underlies the related notions of morphological computation [29–32] and physical intelligence [16, 33], in which sensing, control, and decision-making are partly offloaded from a centralized processor to the mechanical response of the body itself. Solving this inverse problem at low Reynolds number requires a specific approach, which we aim to develop in the present paper.

The natural starting point is the classical mobility theory developed in the 1960s for rigid bodies in Stokes flow [24–26]. Owing to the linearity of the Stokes equations, the six-component velocity of a rigid body depends linearly on the applied force and torque and on the local background velocity, vorticity, and rate-of-strain. The mobility tensors entering this relation depend only on the body's shape and incorporate all the hydrodynamic complexity of the Stokes equations into a finite set of geometric coefficients. These tensors can be computed with different methods:

[*] christophe.eloy@centrale-med.fr

boundary integral methods [34] that exploit Lorentz reciprocal theorem [25], multiblob methods that discretize body surfaces with "blobs" [35–37], or slender-body theory for filaments and fibers [38–40]. This formulation has recently been revived for complex-shaped macromolecules [41–43] and reinterpreted as a low-dimensional dynamical system describing rigid-body sinking trajectories [44, 45]. Its appeal, from the standpoint of the inverse problem, is the explicit, shape-dependent linear relation it provides between velocities and loads [46].

For N rigid bodies, the same linear relationship extends to a $6N \times 6N$ grand mobility tensor encoding both single-body mobility and many-body hydrodynamic interactions. This tensor is at the core of Stokesian dynamics, which aims to simulate a large ensemble of rigid bodies interacting in a flow [47–50]. To compute the grand mobility tensor efficiently, multipole expansions are generally used, such as the Rotne–Prager–Yamakawa (RPY) approximation, which provides a closed-form mobility for assemblies of spheres of possibly different radii [42, 51–54]. This approach is now mature for both passive rigid suspensions and active suspensions of rigid swimmers, routinely tackling problems with millions of degrees of freedom [55].

A natural next step has been to leverage these tools to simulate deformable bodies by treating them as assemblies of rigid sub-bodies linked by elastic or rigid constraints. For instance, bead models or multiblob methods approximate filaments as an ensemble of touching spheres connected by springs [15, 56, 57], and general articulated bodies can be simulated in a similar way [36]. At each timestep, however, these formulations require enforcing the constraints, which is a saddle-point linear system, typically solved via preconditioned iterative solvers (GMRES, FMM-accelerated iterations [49, 58]). This is efficient for forward simulation, but would be prohibitive for the inverse problem of shape optimization.

Closest in spirit to the present paper is the framework of Solovev and Friedrich [59]. They explicitly project the deformation of a soft body onto a small set of generalized coordinates using Lagrangian mechanics. This "trick" allows them to compute the dynamics of a shape-changing body without solving the constraints. In the conjugate space of generalized coordinates, they balance generalized hydrodynamic forces against active driving forces, which yields an explicit ordinary differential equation for the generalized velocities. Three differences with the present work should be noted. First, their framework is primarily designed for active microswimmers, not soft bodies deforming elastically. They model active bacterial flagella, eukaryotic cilia, and minimal model swimmers such as Purcell's three-link swimmer [61] or Najafi's three-sphere swimmer [60]. Second, in their formulation, hydrodynamic forces are tabulated with boundary-element methods and interpolated, which makes it difficult to differentiate cleanly for gradient-based optimization. Third, their framework does not include the coupling to the background flow, which is essential for turbulent transport.

The goal of this paper is to develop a general "soft mobility" framework that simultaneously: (1) is derived from the continuum elastohydrodynamic problem of a hyperelastic body in a background Stokes flow; (2) accommodates arbitrary generalized coordinates and yields an explicit, configuration-dependent ordinary differential equation in which mobility, advection, and flow-gradient coupling tensors all appear; and (3) is constructed so that all tensors are smooth, differentiable functions of the design parameters, enabling end-to-end gradient-based inverse design.

The paper is organized as follows. Section II formulates the continuum elastohydrodynamic problem and derives its weak form via the principle of virtual power and the Lorentz reciprocal theorem [25]. Section III specializes the framework to two finite-dimensional reductions — a Galerkin discretization on arbitrary basis modes and a sphere-and-spring assembly — and arrives at the soft mobility equation in closed form, alongside the open-source Python library [62]. Section IV applies the framework to five canonical problems: sinking of a rigid body or a flexible fiber, the instability of a rotating flexible fiber, soft Jeffery orbits, and the optimisation of a three-sphere microswimmer. Finally, we show an example of "smart" navigation through flexibility with the soft surfer. Section V discusses extensions and applications.

II. PROBLEM FORMULATION

We consider a deformable body of characteristic size L immersed in a viscous fluid of kinematic viscosity ν. The Reynolds number is $\mathrm{Re} = UL/\nu$, where U is a characteristic velocity at the body scale. In turbulent environments, the Reynolds number can be calculated by using the Kolmogorov velocity $u_\eta = (\nu\varepsilon)^{1/4}$ as this characteristic velocity, such that $\mathrm{Re} = L/\eta$, with $\eta = (\nu^3/\varepsilon)^{1/4}$ the Kolmogorov length scale. An alternative choice is to estimate the characteristic velocity from the local velocity gradient. In that case, the caracteristic velocity is $U \sim \|\nabla \boldsymbol{u}^\infty\| L \sim L/\tau_\eta$, with $\tau_\eta = (\nu/\varepsilon)^{1/2}$ the Kolmogorov time scale, such that $\mathrm{Re} = (L/\eta)^2$.

Throughout this work, we assume $\mathrm{Re} \ll 1$, so that inertia is negligible. Both choices yield $L \ll \eta$, where η is the Kolmogorov length scale, as the condition for this approximation to hold. Under this assumption, the body is in quasi-static equilibrium at each instant, and the surrounding fluid obeys the Stokes equations.

A further consequence of the small-Reynolds-number assumption is that the background flow $\boldsymbol{u}^\infty(\boldsymbol{r})$ can be lin-

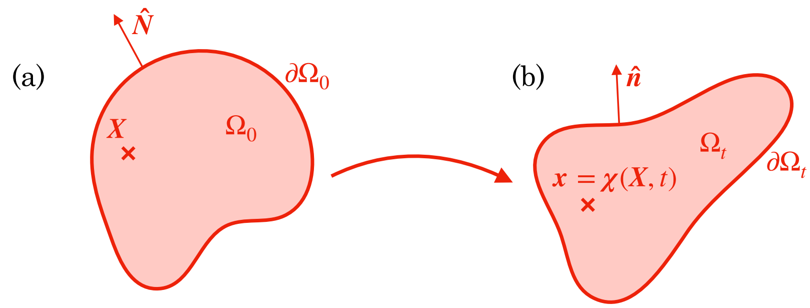

FIG. 1. Representation of the body in (a) the reference configuration and (b) the deformed configuration.

earized about a reference point $\boldsymbol{r}_0$ attached to the body,

$$\boldsymbol{u}^\infty(\boldsymbol{r}) \approx \boldsymbol{u}_0^\infty + \boldsymbol{\omega}_0^\infty \times (\boldsymbol{r} - \boldsymbol{r}_0) + \boldsymbol{E}_0^\infty \cdot (\boldsymbol{r} - \boldsymbol{r}_0), \tag{1}$$

with $\boldsymbol{u}_0^\infty$ the translational velocity, $\boldsymbol{\omega}_0^\infty = \frac{1}{2}\nabla \times \boldsymbol{u}^\infty$ the angular velocity, and $\boldsymbol{E}_0^\infty = \frac{1}{2}(\nabla \boldsymbol{u}^\infty + \nabla^T \boldsymbol{u}^\infty)$ the rate-of-strain tensor of the background flow evaluated at $\boldsymbol{r}_0$.

A. Body

In its undeformed configuration, the body occupies a reference volume Ω_0, with boundary $\partial\Omega_0$. Under the action of the flow and internal stresses, a material point $\boldsymbol{X} \in \Omega_0$ is carried to its current position $\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X}, t) \in \Omega_t$ through the deformation map $\boldsymbol{\chi}$ (Fig. 1). The deformation gradient tensor $\boldsymbol{F} = \partial\boldsymbol{\chi}/\partial\boldsymbol{X}$ characterizes the local stretching and rotation of the material, and $J = \det \boldsymbol{F}$ is its Jacobian. Mass conservation gives $\rho = \rho_0/J$, relating the density ρ in the deformed configuration to its reference value ρ_0.

The material is assumed hyperelastic, so that the second Piola–Kirchhoff stress tensor $\boldsymbol{S}$ derives from a strain energy density $W(\boldsymbol{E})$ as $\boldsymbol{S} = \partial W/\partial \boldsymbol{E}$, where $\boldsymbol{E} = (\boldsymbol{F}^T \cdot \boldsymbol{F} - \boldsymbol{1})/2$ is the Lagrangian Green strain tensor. The outward unit normals to the body surface in the reference and deformed configurations are denoted $\boldsymbol{N}(\boldsymbol{X})$ and $\boldsymbol{n}(\boldsymbol{x})$, respectively. The first Piola–Kirchhoff stress tensor $\boldsymbol{P} = \boldsymbol{F} \cdot \boldsymbol{S}$ provides the traction $\boldsymbol{F}_s$ in the reference configuration: $\boldsymbol{F}_s = \boldsymbol{P} \cdot \boldsymbol{N}$ is the force per unit reference area exerted on a surface element with outward normal $\boldsymbol{N}$, so that the surrounding fluid exerts a force $\boldsymbol{F}_s$ on the body.

Our goal is to determine the velocity field $\dot{\boldsymbol{\chi}}$ of the body in terms of the applied forces and its elastic properties. At low Reynolds number, inertia is negligible and the body is in quasi-static equilibrium at each instant. The equilibrium equations in the reference configuration are

$$\nabla_{\boldsymbol{X}} \cdot \boldsymbol{P} + \boldsymbol{F}_v = 0, \quad \text{in } \Omega_0, \tag{2}$$

$$\boldsymbol{P} \cdot \boldsymbol{N} = \boldsymbol{F}_s, \quad \text{on } \partial\Omega_0, \tag{3}$$

expressing the balance of body forces $\boldsymbol{F}_v$ against the divergence of the first Piola–Kirchhoff stress, and the continuity of traction at the solid–fluid interface.

To obtain a weak formulation of the equilibrium equations, we apply the principle of virtual power: for any kinematically admissible virtual velocity field $\boldsymbol{V}^*(\boldsymbol{X})$, the internal virtual power equals the power of the external forces

$$\int_{\Omega_0} \boldsymbol{S} : \dot{\boldsymbol{E}}^* \, \mathrm{d}V = \int_{\Omega_0} \boldsymbol{F}_v \cdot \boldsymbol{V}^* \, \mathrm{d}V + \int_{\partial\Omega_0} \boldsymbol{F}_s \cdot \boldsymbol{V}^* \, \mathrm{d}S, \tag{4}$$

where the virtual strain rate is

$$\dot{\boldsymbol{E}}^* = \mathrm{sym}\left(\boldsymbol{F}^T \cdot \nabla \boldsymbol{V}^*\right). \tag{5}$$

We assume that the body force is linear in an externally applied field $\boldsymbol{H}$, uniform at the scale of the body,

$$\boldsymbol{F}_v(\boldsymbol{x}) = \rho(\boldsymbol{x})\, \boldsymbol{C}_H(\boldsymbol{x}) \cdot \boldsymbol{H} = \rho_0(\boldsymbol{X})\, \boldsymbol{F} \cdot \boldsymbol{C}_H(\boldsymbol{X}) \cdot \boldsymbol{F}^{-1} \cdot \boldsymbol{H}, \tag{6}$$

where $\boldsymbol{C}_H(\boldsymbol{x})$ is a configuration-dependent coupling tensor. The second equality uses $\rho = \rho_0/J$ and the pushforward of $\boldsymbol{C}_H$ to the reference configuration. This parameterization covers gravity ($\boldsymbol{H} = \boldsymbol{g}$, $\boldsymbol{C}_H = \boldsymbol{1}$) and magnetic forces ($\boldsymbol{H} = \boldsymbol{B}$, $\boldsymbol{C}_H = [\boldsymbol{\mu}]_\times$, where $\boldsymbol{\mu}$ is the magnetic moment per unit mass).

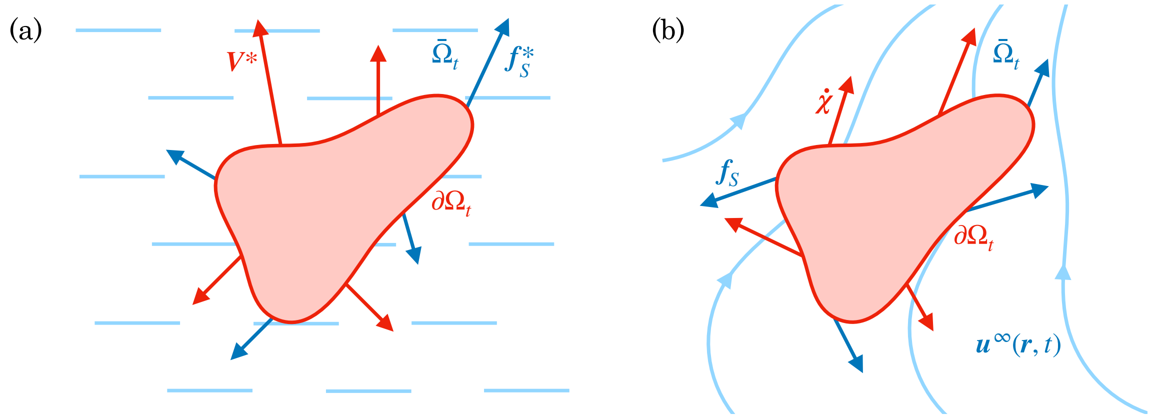

FIG. 2. Two solutions of the Stokes equations used in the reciprocal theorem: (a) the auxiliary problem, driven by virtual surface velocity $\boldsymbol{V}^*$ in an otherwise quiescent fluid, producing traction $\boldsymbol{f}_s^*$ on the left-hand side of Eq. (11); (b) the physical problem, the body moving with velocity $\dot{\boldsymbol{\chi}}$ in background flow $\boldsymbol{u}^\infty$, producing traction $\boldsymbol{f}_s$ on the right-hand side of Eq. (11).

B. Fluid

The fluid occupies the exterior domain $\bar{\Omega}_t$ and its motion satisfies the Stokes equations,

$$\nabla p = \mu \nabla^2 \boldsymbol{u}, \quad \text{in } \bar{\Omega}_t, \tag{7}$$

$$\nabla \cdot \boldsymbol{u} = 0, \quad \text{in } \bar{\Omega}_t. \tag{8}$$

We introduce two Stokes solutions that share the same geometry but differ in their boundary conditions on $\partial\Omega_t$ (Fig. 2). The physical solution $(\boldsymbol{u}, p)$ represents the disturbance flow due to the moving body; the total fluid velocity is $\boldsymbol{u} + \boldsymbol{u}^\infty$, with boundary condition

$$\boldsymbol{u} = \dot{\boldsymbol{\chi}} - \boldsymbol{u}^\infty \quad \text{on } \partial\Omega_t. \tag{9}$$

The auxiliary solution $(\boldsymbol{u}^*, p^*)$ is driven by the virtual velocity field,

$$\boldsymbol{u}^* = \boldsymbol{V}^* \quad \text{on } \partial\Omega_t. \tag{10}$$

Each solution produces a traction on the body surface: $\boldsymbol{f}_s$ for the physical flow and $\boldsymbol{f}_s^*$ for the virtual flow. Applying the Lorentz reciprocal theorem [25] to these two Stokes flows gives

$$\int_{\partial\Omega_t} \boldsymbol{f}_s \cdot \boldsymbol{V}^* \, \mathrm{d}s = \int_{\partial\Omega_t} \boldsymbol{f}_s^* \cdot (\dot{\boldsymbol{\chi}} - \boldsymbol{u}^\infty) \, \mathrm{d}s. \tag{11}$$

These integrals in the deformed configuration can also be expressed in the reference configuration using the surface Jacobian J_s (Eq. A2)

$$\int_{\partial\Omega_t} \boldsymbol{f}_s \cdot \boldsymbol{V}^* \, \mathrm{d}s = \int_{\partial\Omega_0} \boldsymbol{F}_s \cdot \boldsymbol{V}^* \, \mathrm{d}S, \tag{12}$$

with $\boldsymbol{F}_s = J_s \boldsymbol{f}_s$ (and similarly $\boldsymbol{F}_s^* = J_s \boldsymbol{f}_s^*$) the surface force in the reference configuration. We recognize here the last term of Eq. (4), which can thus conveniently be calculated through the following equality

$$\int_{\partial\Omega_0} \boldsymbol{F}_s \cdot \boldsymbol{V}^* \, \mathrm{d}S = \int_{\partial\Omega_0} \boldsymbol{F}_s^* \cdot (\dot{\boldsymbol{\chi}} - \boldsymbol{u}^\infty) \, \mathrm{d}S. \tag{13}$$

But to do that, we need to calculate the virtual traction $\boldsymbol{F}_s^* = J_s \boldsymbol{f}_s^*$ for a body of shape $\boldsymbol{x}$ with surface velocity $\boldsymbol{V}^*$. From the linearity of the Stokes equations, we know that the relation between $\boldsymbol{f}_s^*$ and $\boldsymbol{V}^*$ is linear, but depends nonlinearly on the shape $\boldsymbol{x}$. In general, $\boldsymbol{F}_s^*$ can be computed through the boundary integral equation (Appendix A 2).

C. Summary

Using Eqs. (4), (11), and (13), we find the weak formulation of the problem on the space V of kinematically admissible velocity fields: one seeks $\dot{\boldsymbol{\chi}} \in V$ satisfying

$$\int_{\partial\Omega_0} \boldsymbol{F}_s^* \cdot \dot{\boldsymbol{\chi}} \, \mathrm{d}S = \int_{\partial\Omega_0} \boldsymbol{F}_s^* \cdot \boldsymbol{u}^\infty \, \mathrm{d}S + \int_{\Omega_0} \boldsymbol{S} : \dot{\boldsymbol{E}}^* \, \mathrm{d}V - \int_{\Omega_0} \boldsymbol{F}_v \cdot \boldsymbol{V}^* \, \mathrm{d}V, \tag{14}$$

for all virtual velocities $\boldsymbol{V}^* \in V$.

The left-hand side is linear in the unknown $\dot{\boldsymbol{\chi}}$ but nonlinear in the body's shape. The right-hand side is independent of $\dot{\boldsymbol{\chi}}$ and is composed of three terms: (1) the virtual power of the viscous forces due to the background flow, (2) the virtual elastic power, and (3) the virtual power of the body forces.

The structure of Eq. (14) becomes transparent through the decomposition of $\dot{\boldsymbol{\chi}}$ into a rigid-body part and a strain-producing deformation part. Let $\boldsymbol{u}_0$, $\boldsymbol{\omega}_0$ be the translational and rotational velocity of the body frame. Then

$$\dot{\boldsymbol{\chi}}(\boldsymbol{X}, t) = \boldsymbol{u}_0 + \boldsymbol{\omega}_0 \times (\boldsymbol{\chi}(\boldsymbol{x}, t) - \boldsymbol{r}_0) + \boldsymbol{V}_{\mathrm{def}}(\boldsymbol{X}, t), \tag{15}$$

where $\boldsymbol{U}(\boldsymbol{r})$ is the transport tensor of Eq. (1) and $\boldsymbol{V}_{\mathrm{def}}$ is the velocity due to shape change at fixed $\boldsymbol{v}_0$.

Two structural consequences follow. First, for any rigid-body virtual velocity $\boldsymbol{V}^* = \boldsymbol{U}(\boldsymbol{\chi} - \boldsymbol{r}_0) \cdot \boldsymbol{v}_0^*$, the hyperelastic term in Eq. (14) vanishes. This is because $\boldsymbol{F}^T \cdot \nabla \boldsymbol{V}^*$ is then skew-symmetric in Eq. (5) when $\nabla \boldsymbol{V}^* = [\boldsymbol{\omega}_0^*]_\times \cdot \boldsymbol{F}$. Second, comparing Eq. (15) with the background flow linearization (1), the same transport operator $\boldsymbol{U}(\boldsymbol{\chi} - \boldsymbol{r}_0)$ couples $\boldsymbol{v}_0$ on the left of Eq. (14) and $\boldsymbol{v}_0^\infty$ on the right. Only the relative velocity $\boldsymbol{v}_0 - \boldsymbol{v}_0^\infty$ therefore drives the hydrodynamic coupling, consistent with Galilean invariance.

III. NUMERICAL APPROACHES

A. Galerkin discretization

Equation (14) is a weak formulation on infinite-dimensional space V. Because the background flow $\boldsymbol{u}^\infty$ is linear in $\boldsymbol{v}_0^\infty = [\boldsymbol{u}_0^\infty, \boldsymbol{\omega}_0^\infty]$ and $\boldsymbol{E}_0^\infty$ [Eq. (1)], and the problem is linear in $\dot{\boldsymbol{\chi}}$, the solution decomposes into contributions driven by each input separately.

To obtain a finite-dimensional system, we restrict $\dot{\boldsymbol{\chi}}$ to a subspace $V_n \subset V$ spanned by n basis functions $\{\boldsymbol{V}_j\}$ and write

$$\dot{\boldsymbol{\chi}}(\boldsymbol{X}, t) = \sum_{j=1}^{n} \dot{a}_j(t) \, \boldsymbol{V}_j(\boldsymbol{X}). \tag{16}$$

Testing Eq. (14) with each $\boldsymbol{V}_i$ and using the linearity in $\boldsymbol{v}_0^\infty$ and $\boldsymbol{E}_0^\infty$ yields the projected system

$$\hat{\boldsymbol{R}} \cdot \dot{\boldsymbol{a}} + \hat{\boldsymbol{K}}_{NL} - \hat{\boldsymbol{C}}_H \cdot \boldsymbol{H} + \hat{\boldsymbol{C}}_V \cdot \boldsymbol{v}_0^\infty + \hat{\boldsymbol{C}}_E : \boldsymbol{E}_0^\infty = 0, \tag{17}$$

where $\boldsymbol{a} = [a_1, \ldots a_n]$ and $\boldsymbol{E}_0^\infty$ is the 3×3 symmetric traceless rate-of-strain tensor, which in practice is represented as a 5-component vector (see Appendix B 2). All tensors depend on the current configuration $\boldsymbol{\chi}(\boldsymbol{X}, t)$. The resistance matrix $\hat{\boldsymbol{R}}$ is symmetric positive definite by virtue of the reciprocal theorem. The nonlinear stiffness $\hat{\boldsymbol{K}}_{NL}$ is symmetric positive semi-definite by its relation to the elastic energy.

Inverting Eq. (17) allows one to compute $\dot{\boldsymbol{a}}$ and thus to simulate through time integration the transport and deformation of a soft body in a background flow. The difficulty is that all tensors in Eq. (17) depend on the current shape through the vector $\boldsymbol{a}$. An efficient method would be to project the deformation onto the most meaningful modes V_n, for instance, by computing the least attenuated viscous eigenmodes of the system with no background flow and no body forces. An alternative is to consider a soft body made of rigid spheres connected by springs, as done below.

B. Kinematics of a sphere assembly

The general Galerkin framework of Sec. III.A applies to any compatible velocity basis $\{\boldsymbol{V}_j\}$. For bodies composed of rigid spheres connected by elastic links, the natural choice is the rigid-body modes of each sphere. The sphere-assembly discretization developed here is the specialization of the Galerkin method to this basis.

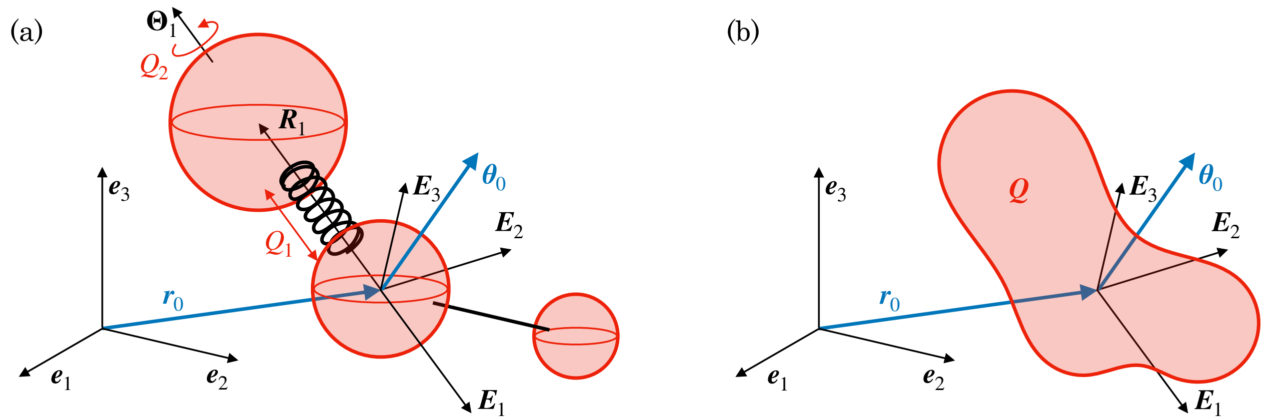

FIG. 3. Notations for an assembly of spheres. (a) Assembly of $N = 3$ spheres with $N_Q = 2$ degrees of freedom $\boldsymbol{Q} = [Q_1, Q_2]$. In the assembly frame, the i-th sphere is at position $\boldsymbol{R}_i$ and has an orientation $\boldsymbol{\Theta}_i$. The assembly frame is at position $\boldsymbol{r}_0$ and orientation $\boldsymbol{\theta}_0$ in the laboratory frame. (b) The assembly configuration is entirely represented by its position $\boldsymbol{r}_0$, its orientation $\boldsymbol{\theta}_0$, and its degrees of freedom $\boldsymbol{Q}$.

We consider an assembly of N spheres connected by springs and rods. The i-th sphere of radius a_i has center position $\boldsymbol{R}_i$ and Rodrigues orientation vector $\boldsymbol{\Theta}_i$ in the body frame $(\boldsymbol{E}_1, \boldsymbol{E}_2, \boldsymbol{E}_3)$. Both are gathered in a six-component position $\boldsymbol{X}_i(\boldsymbol{Q}, t)$, a function of time and the N_Q degrees of freedom $\boldsymbol{Q}$ representing the deformation (Fig. 3). The associated six-component velocity is $\boldsymbol{V}_i = [\boldsymbol{U}_i, \boldsymbol{\Omega}_i]$, with $\boldsymbol{U}_i$ the translational velocity and $\boldsymbol{\Omega}_i$ the angular velocity. The six-component velocity $\boldsymbol{V}_i$ is related to $\dot{\boldsymbol{Q}}$ by the per-sphere Jacobian $\boldsymbol{J}_i = \partial \boldsymbol{V}_i / \partial \dot{\boldsymbol{Q}}$

$$\boldsymbol{V}_i = \boldsymbol{J}_i \cdot \dot{\boldsymbol{Q}} + (\boldsymbol{V}_{\mathrm{act}})_i, \tag{18}$$

with $(\boldsymbol{V}_{\mathrm{act}})_i = \boldsymbol{B}_i \cdot \partial \boldsymbol{X}_i / \partial t$ an active component due to prescibed kinematics and $\boldsymbol{B}_i(\boldsymbol{\Theta}_i)$ the Bortz operator (see Appendix B 1).

Stacking all spheres, the grand velocity $\boldsymbol{V} = [\boldsymbol{V}_1, \ldots, \boldsymbol{V}_N]$ in the body frame satisfies

$$\boldsymbol{V} = \boldsymbol{J}_{\mathrm{sph}} \cdot \dot{\boldsymbol{Q}} + \boldsymbol{V}_{\mathrm{act}}, \tag{19}$$

with the grand active velocity $\boldsymbol{V}_{\mathrm{act}} = [(\boldsymbol{V}_{\mathrm{act}})_1, \ldots, (\boldsymbol{V}_{\mathrm{act}})_N]$ of size $6N$ collecting the per-sphere contributions, and the assembly Jacobian $\boldsymbol{J}_{\mathrm{sph}}(\boldsymbol{Q}, t) = \partial \boldsymbol{V} / \partial \dot{\boldsymbol{Q}}$ of size $6N \times N_Q$. Because $\boldsymbol{V}$ is affine in $\dot{\boldsymbol{Q}}$, it is a quasi-velocity of the Lagrangian system.

The body's frame $(\boldsymbol{E}_1, \boldsymbol{E}_2, \boldsymbol{E}_3)$ undergoes a global rigid-body motion. Its position and orientation are encoded in the six-component position $\boldsymbol{x}_0 = [\boldsymbol{r}_0, \boldsymbol{\theta}_0]$, where $\boldsymbol{r}_0$ is the position and $\boldsymbol{\theta}_0$ the Rodrigues orientation of the body's frame. The full state of the assembly in the laboratory frame is therefore given by the generalized coordinates (of size $N_q = 6 + N_Q$)

$$\boldsymbol{q} = [\boldsymbol{x}_0,\, \boldsymbol{Q}]\,. \tag{20}$$

We also define the six-component velocity $\boldsymbol{v}_0 = [\boldsymbol{u}_0, \boldsymbol{\omega}_0]$, with $\boldsymbol{u}_0$ the translational velocity and $\boldsymbol{\omega}_0$ the angular velocity of the body in the laboratory frame. This six-component velocity is associated with a generalized velocity

$$\boldsymbol{p} = \left[\boldsymbol{u}_0,\, \boldsymbol{\omega}_0,\, \dot{\boldsymbol{Q}}\right] = \boldsymbol{B} \cdot \dot{\boldsymbol{q}}, \tag{21}$$

where $\boldsymbol{B}(\boldsymbol{\theta}_0)$ is a Bortz matrix given inAppendix B 1.

The laboratory-frame velocity of sphere i combines the body-frame velocity $\boldsymbol{V}_i$ with the rigid-body transport due to $\boldsymbol{v}_0$:

$$\boldsymbol{u}_i = \boldsymbol{U}_i + \boldsymbol{u}_0 + \boldsymbol{\omega}_0 \times \boldsymbol{R}_i, \quad \boldsymbol{\omega}_i = \boldsymbol{\Omega}_i + \boldsymbol{\omega}_0.$$

Stacking all spheres, this composition takes the compact form

$$\boldsymbol{v} = \boldsymbol{V} + \boldsymbol{C}_U \cdot \boldsymbol{v}_0 = \boldsymbol{V} + \boldsymbol{C}_U \cdot \boldsymbol{B}_0 \cdot \dot{\boldsymbol{x}}_0, \tag{22}$$

with $\boldsymbol{B}_0 = \boldsymbol{B}_i(\boldsymbol{\theta}_0)$ the Bortz operator evaluated at the body-frame orientation and $\boldsymbol{C}_U$ a $6N \times 6$ transport coupling tensor. Combining Eqs. (19) and (22), the grand velocity of all spheres in the laboratory frame is related to $\boldsymbol{p}$ by

$$\boldsymbol{v} = \boldsymbol{V}_{\text{act}} + \boldsymbol{J} \cdot \boldsymbol{p} = \boldsymbol{V}_{\text{act}} + \boldsymbol{J} \cdot \boldsymbol{B} \cdot \dot{\boldsymbol{q}}, \tag{23}$$

where $\boldsymbol{J} = \partial \boldsymbol{v}/\partial \boldsymbol{p}$ is the $6N \times N_q$ full Jacobian. Because $\boldsymbol{B}$ is invertible, the grand sphere velocities $\boldsymbol{v}$ are quasi-velocities of this Lagrangian system.

The linearized background flow [Eq. (1)] evaluated at the sphere centers gives the grand flow velocity $\boldsymbol{v}^\infty = [\boldsymbol{v}_1^\infty, \dots, \boldsymbol{v}_N^\infty]$ and the rate-of-strain tensor

$$\boldsymbol{v}^\infty = \boldsymbol{C}_U \cdot \boldsymbol{v}_0^\infty + \boldsymbol{C}_S : \boldsymbol{E}_0^\infty, \quad \boldsymbol{E}^\infty = \boldsymbol{E}_0^\infty, \tag{24}$$

where $\boldsymbol{C}_U$ is the same transport coupling tensor as in Eq. (22) and $\boldsymbol{C}_S$ is a $6N \times 3 \times 3$ stresslet coupling tensor. Here $\boldsymbol{v}_0^\infty = [\boldsymbol{u}_0^\infty, \boldsymbol{\omega}_0^\infty]$ denotes the six-component velocity of the undisturbed flow evaluated at the reference point $\boldsymbol{r}_0$. For simplicity, we also introduce the generalized flow velocity $\boldsymbol{p}_0^\infty = [\boldsymbol{u}_0^\infty, \boldsymbol{\omega}_0^\infty, \boldsymbol{0}]$ of size $N_q = 6 + N_Q$.

C. Soft mobility equation of a sphere assembly

With the kinematic description established, we now derive the equation of motion by applying the principle of virtual power to the sphere assembly. We then show how to reduce it to the soft mobility equation.

The six-component force $\boldsymbol{f}_i = [\boldsymbol{F}_i, \boldsymbol{T}_i]$ acting on sphere i collects the force $\boldsymbol{F}_i$ applied at its center and the associated torque $\boldsymbol{T}_i$. Stacking all forces into the grand force $\boldsymbol{f} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_N]$, it can be decomposed as

$$\boldsymbol{f} = \boldsymbol{f}_c + \boldsymbol{f}_{\text{hydro}} + \boldsymbol{f}_{\text{ext}} + \boldsymbol{f}_K,$$

where $\boldsymbol{f}_c$ are internal constraint forces (maintaining prescribed inter-sphere constraints), $\boldsymbol{f}_{\text{hydro}}$ are hydrodynamic forces, $\boldsymbol{f}_{\text{ext}}$ are external body forces, and $\boldsymbol{f}_K$ are elastic restoring forces. The elastic forces are linear in the deformation state,

$$\boldsymbol{f}_K = \boldsymbol{C}_K \cdot \boldsymbol{Q}, \tag{25}$$

and satisfy $\sum_{i=1}^N (\boldsymbol{f}_K)_i = 0$ (Newton's third law for internal forces). The body forces are linear in an external field $\boldsymbol{H}$,

$$\boldsymbol{f}_{\text{ext}} = \boldsymbol{C}_H \cdot \boldsymbol{H}. \tag{26}$$

The principle of virtual power allows us to eliminate the unknown constraint forces. For any virtual velocity $\boldsymbol{v}^*$ compatible with the constraints, the constraint forces do not work and hence $\boldsymbol{f}_c \cdot \boldsymbol{v}^* = 0$. The virtual power of all forces can thus be written:

$$P^* = (\boldsymbol{f}_{\text{hydro}} + \boldsymbol{f}_{\text{ext}} + \boldsymbol{f}_K) \cdot \boldsymbol{v}^* = 0, \tag{27}$$

and it vanishes because the forces balance. Since $\boldsymbol{v}^* = (\partial \boldsymbol{v}/\partial \boldsymbol{p}) \cdot \boldsymbol{p}^*$, and the above equality must hold for all $\boldsymbol{p}^*$, one obtains

$$\boldsymbol{J}^T \cdot (\boldsymbol{f}_{\text{hydro}} + \boldsymbol{f}_{\text{ext}} + \boldsymbol{f}_K) = 0, \tag{28}$$

where $\boldsymbol{J} = \partial \boldsymbol{v}/\partial \boldsymbol{p}$ is the same Jacobian as in Eq. (23). This projection, deduced from classical tools of Lagrangian mechanics, reduces the $6N$-dimensional force balance to the N_q dimensions of the generalized coordinates while eliminating the constraint forces $\boldsymbol{f}_c$ from the equations of motion.

The hydrodynamic force on the sphere assembly is given by the grand resistance relation

$$\boldsymbol{f}_{\text{hydro}} = \boldsymbol{R} \cdot (\boldsymbol{v}^\infty - \boldsymbol{v}) + \boldsymbol{R}_S : \boldsymbol{E}^\infty, \tag{29}$$

where $\boldsymbol{R}$ is the $6N \times 6N$ grand resistance matrix and $\boldsymbol{R}_S$ is the $6N \times 3 \times 3$ stresslet resistance tensor. Substituting Eqs. (23), (24), and (29) into the generalized force balance (28), and writing $\boldsymbol{f} = \boldsymbol{f}_{\text{ext}} + \boldsymbol{f}_K$ for the non-hydrodynamic forces, one obtains

$$\begin{aligned}
\boldsymbol{J}^T \cdot \boldsymbol{R} \cdot \boldsymbol{v} &= \boldsymbol{J}^T \cdot [\boldsymbol{f} + \boldsymbol{R} \cdot \boldsymbol{v}^\infty + \boldsymbol{R}_S : \boldsymbol{E}^\infty], && (30)\\
\boldsymbol{p} &= \boldsymbol{\Pi} \cdot \left[\boldsymbol{R}^{-1} \cdot \boldsymbol{f} - \boldsymbol{V}_{\text{act}} + \boldsymbol{C}_U \cdot \boldsymbol{v}_0^\infty + (\boldsymbol{C}_S + \boldsymbol{R}^{-1} \cdot \boldsymbol{R}_S) : \boldsymbol{E}_0^\infty\right], && (31)\\
\boldsymbol{p} &= \boldsymbol{M} \cdot \boldsymbol{f} - \boldsymbol{\Pi} \cdot \boldsymbol{V}_{\text{act}} + \boldsymbol{p}_0^\infty + \boldsymbol{C}_E : \boldsymbol{E}_0^\infty, && (32)
\end{aligned}$$

upon introducing the $N_q \times 6N$ projection operator

$$\boldsymbol{\Pi} = \left(\boldsymbol{J}^T \cdot \boldsymbol{R} \cdot \boldsymbol{J}\right)^{-1} \cdot \boldsymbol{J}^T \cdot \boldsymbol{R}, \tag{33}$$

which satisfies $\boldsymbol{\Pi} \cdot \boldsymbol{J} = \boldsymbol{1}$ and eliminates the constraint forces. The soft mobility tensor and reduced flow-coupling tensors are

$$\boldsymbol{M} = \boldsymbol{\Pi} \cdot \boldsymbol{R}^{-1} = \left(\boldsymbol{J}^T \cdot \boldsymbol{R} \cdot \boldsymbol{J}\right)^{-1} \cdot \boldsymbol{J}^T, \quad \boldsymbol{C}_E = \boldsymbol{\Pi} \cdot (\boldsymbol{C}_S + \boldsymbol{R}^{-1} \cdot \boldsymbol{R}_S). \tag{34}$$

The tensor $\boldsymbol{M}$ ($N_q \times 6N$) extends the classical rigid-body mobility to deformable assemblies. Substituting the force decompositions (25)–(26) yields

$$\boldsymbol{M} \cdot \boldsymbol{f}_K = \boldsymbol{M}_K \cdot \boldsymbol{Q}, \quad \boldsymbol{M} \cdot \boldsymbol{f}_{\text{ext}} = \boldsymbol{M}_H \cdot \boldsymbol{H}, \tag{35}$$

with $\boldsymbol{M}_K = \boldsymbol{M} \cdot \boldsymbol{C}_K$ of size $N_q \times N_Q$ and $\boldsymbol{M}_H = \boldsymbol{M} \cdot \boldsymbol{C}_H$ of size $N_q \times 3$.

Equation (32) constitutes the soft mobility equation of the assembly. Expressed in the body's frame, it reads

$$\boldsymbol{p} - \boldsymbol{p}_0^\infty = \begin{pmatrix} \boldsymbol{u}_0 - \boldsymbol{u}_0^\infty \\ \boldsymbol{\omega}_0 - \boldsymbol{\omega}_0^\infty \\ \dot{\boldsymbol{Q}} \end{pmatrix} = \boldsymbol{M}_K \cdot \boldsymbol{Q} + \boldsymbol{M}_H \cdot \boldsymbol{H} + \boldsymbol{C}_E : \boldsymbol{E}_0^\infty - \boldsymbol{\Pi} \cdot \boldsymbol{V}_{\text{act}}, \tag{36}$$

where $\boldsymbol{M}_K$, $\boldsymbol{M}_H$, $\boldsymbol{C}_E$, and $\boldsymbol{\Pi}$ depend only on the deformation state $\boldsymbol{Q}$. Since $\boldsymbol{p} = \boldsymbol{B}^{-1} \cdot \dot{\boldsymbol{q}}$, this is a first-order ODE of the form $\dot{\boldsymbol{q}} = f(\boldsymbol{q}, t)$, which can be integrated forward in time given initial conditions and the prescribed flow and external fields. When the flow and external fields are known in the laboratory frame, they must be rotated into the body's frame before evaluating the right-hand side:

$$\boldsymbol{u}_0^\infty = \boldsymbol{\mathcal{R}}_0^T \cdot (\boldsymbol{u}_0^\infty)_{\text{lab}}\,, \quad \boldsymbol{E}_0^\infty = \boldsymbol{\mathcal{R}}_0^T \cdot (\boldsymbol{E}_0^\infty)_{\text{lab}} \cdot \boldsymbol{\mathcal{R}}_0, \quad \boldsymbol{H} = \boldsymbol{\mathcal{R}}_0^T \cdot (\boldsymbol{H})_{\text{lab}}\,, \tag{37}$$

where $\boldsymbol{\mathcal{R}}_0(\boldsymbol{\theta}_0)$ is the rotation matrix given by the Euler–Rodrigues formula (B1).

D. Python library

The soft mobility model for sphere assemblies described above is implemented in a Python library, freely available as open-source code [62]. The reader is encouraged to test it, all results presented below are available as tutorial notebooks in the library.

The soft mobility library allows the user to define a sphere assembly and to integrate its trajectory in a flow. The main workflow proceeds in five steps: (1) define the sphere-assembly characteristics and create an instance of the `SoftBody` class; (2) compute the soft mobility tensors of Eq. (36); (3) specify the background flow, the external fields associated with body forces, and the kinematic or internal forces laws; (4) integrate the trajectory with the class `FlowBodyRollout`; and (5) optionally optimize design variables with the class `FlowBodyOptimizer`.

Each soft body partitions its parameters into three groups. First, degrees of freedom (`dofs`) are the generalized coordinates $\boldsymbol{Q}$ associated with deformation, such as spring extensions or hinge angles, that evolve during a simulation. Second, design variables (`design`) are morphological and material parameters, such as sphere radii, rest lengths, and stiffnesses. They are fixed during a given simulation and are the natural targets for optimization. Third, inputs (`inputs`) are external or active controls (such as a gravity field or an imposed kinematic deformation). Three-component variable names ending in `0`, `1`, `2` (e.g. `gravity0...2`) are automatically treated as vector fields, while single-component names are treated as scalar controls. The distinction is made because fields rotate when changing the reference frame through Eq. (37) and scalars do not

Sphere assemblies are most conveniently defined via a compact YAML description in which the position, orientation, force, and torque of each sphere are written as symbolic expressions in the declared variable names. For instance, this is a YAML file used to model a dumbbell composed of two spheres of equal radius, where the design variables are the stiffness and rest length of the spring:

```
dof_names:    [elongation]
design_names: [rest_length, stiffness]
defaults:
  radius: 1.0
  rest_length: 1.0
```

```
  stiffness: 1.0
spheres:
  - radius: radius
    position: [-radius - rest_length, 0, 0]
    force: [stiffness * elongation, 0, 0]
  - radius: radius
    position: [radius + elongation, 0, 0]
    force: [-stiffness * elongation, 0, 0]
```

The parser automatically classifies each symbol into its variable group, and linearizes the force and torque expressions in the inputs to construct the coupling matrices $\boldsymbol{C}_K$ and $\boldsymbol{C}_H$ introduced in Eqs. (25) and (26). The YAML string is passed directly to `SoftBody`, which builds the sphere assembly and computes all soft mobility tensors in Eq. (36).

Input objects are represented by three classes: (1) A `Flow` object provides the background fluid velocity $\boldsymbol{u}_0^\infty$, its gradient $\nabla \boldsymbol{u}^\infty$, and the decomposition into vorticity $\boldsymbol{\omega}_0^\infty$ and rate-of-strain $\boldsymbol{E}_0^\infty$ required by Eq. (36); (2) A `Field` object returns a time- and position-dependent three-component vector (e.g. gravity $\boldsymbol{g}$ or a rotating magnetic field $\boldsymbol{B}$) and is coupled to the body through the external-field term $\boldsymbol{H}$. (3) A `Scalar` object returns a single real value as a function of position and time, suitable for prescribed active controls or time-dependent coefficients.

All numerical operations are implemented with JAX [63], enabling just-in-time compilation (`jax.jit`) for performance and automatic differentiation (`jax.grad`) through the entire simulation. The latter is exploited by `FlowBodyOptimizer`, which pairs the differentiable rollout with gradient-based optimizers from the Optax library [64] to minimize or maximize a user-defined scalar objective of the trajectory with respect to the design variables.

IV. RESULTS

A. Sinking of a rigid body

B. Sinking of a flexible fiber

C. Instability of a rotating flexible fiber

D. Jeffery orbits for rigid and soft bodies

E. Optimisation of a three-sphere microswimmer

F. Soft surfer

V. DISCUSSION

Appendix A: Continuum mechanics

1. Surface Jacobian

The surface integral can also be expressed in the deformed configuration

$$\int_{\partial\Omega_0} \boldsymbol{F}_s \cdot \boldsymbol{V}^* \, \mathrm{d}S = \int_{\partial\Omega_t} \boldsymbol{f}_s \cdot \boldsymbol{V}^* \, \mathrm{d}s = \int_{\partial\Omega_0} J_s \boldsymbol{f}_s \cdot \boldsymbol{V}^* \, \mathrm{d}S, \tag{A1}$$

with $\boldsymbol{f}_s = \boldsymbol{\sigma} \cdot \boldsymbol{n}$ the traction (in the deformed configuration), the force exerted by the fluid on the body, where $\boldsymbol{\sigma}$ is the Cauchy stress tensor. We thus have $\boldsymbol{F}_s = J_s \boldsymbol{f}_s$ with J_s the Jacobian of the surface deformation

$$J_s = \|\boldsymbol{F}^{-T} \cdot \boldsymbol{N}\| \det(\boldsymbol{F}). \tag{A2}$$

2. Boundary integral equation

The hydrodynamic traction $\boldsymbol{f}_s(\boldsymbol{z})$ at a point $\boldsymbol{z} \in \partial\Omega_t$ is determined by the boundary integral formulation of the Stokes equations [34]. By linearity, the surface traction and the surface velocity are related through

$$\boldsymbol{u}(\boldsymbol{z}) \;=\; -\frac{1}{4\pi\mu}\int_{\partial\Omega_t} \boldsymbol{G}(\boldsymbol{y}-\boldsymbol{z})\cdot\boldsymbol{f}_s(\boldsymbol{y})\,\mathrm{d}s(\boldsymbol{y}) + \frac{1}{4\pi}\int_{\partial\Omega_t} \boldsymbol{u}(\boldsymbol{y})\cdot\boldsymbol{T}(\boldsymbol{y}-\boldsymbol{z})\cdot\boldsymbol{n}(\boldsymbol{y})\,\mathrm{d}s(\boldsymbol{y}), \tag{A3}$$

$$\;=\; -\frac{1}{8\pi\mu}\int_{\partial\Omega_t} \boldsymbol{G}(\boldsymbol{y}-\boldsymbol{z})\cdot\boldsymbol{f}_s(\boldsymbol{y})\,\mathrm{d}s(\boldsymbol{y}) + \frac{1}{8\pi}\int_{\partial\Omega_t} [\boldsymbol{u}(\boldsymbol{y})-\boldsymbol{u}(\boldsymbol{z})]\cdot\boldsymbol{T}(\boldsymbol{y}-\boldsymbol{z})\cdot\boldsymbol{n}(\boldsymbol{y})\,\mathrm{d}s(\boldsymbol{y}), \tag{A4}$$

with

$$\boldsymbol{G}(\boldsymbol{r}) \;=\; \frac{\boldsymbol{1}}{r} + \frac{\boldsymbol{r}\otimes\boldsymbol{r}}{r^3}, \tag{A5}$$

$$\boldsymbol{T}(\boldsymbol{r}) \;=\; -6\frac{\boldsymbol{r}\otimes\boldsymbol{r}\otimes\boldsymbol{r}}{r^5}, \tag{A6}$$

where $\boldsymbol{G}$ is the second-order stokeslet Oseen tensor, $\boldsymbol{T}$, the corresponding third-order stress tensor, and $r = \|\boldsymbol{r}\|$. Both Eqs. (A3) and (A4) are equivalent. In Eq. (A4), subtracting $\boldsymbol{u}(\boldsymbol{z})$ from $\boldsymbol{u}(\boldsymbol{y})$ in the double-layer integrand cancels the $1/r^2$ singularity of $\boldsymbol{T}$, so the integral is weakly singular and standard Gauss quadrature applies.

Setting $\boldsymbol{u}(\boldsymbol{z}) = \boldsymbol{V}^*(\boldsymbol{z})$ on $\partial\Omega_t$ in Eq. (A4) yields the virtual traction $\boldsymbol{f}_s^*$ required in the reciprocal theorem (11).

Appendix B: Sphere assembly kinematics

1. Bortz equation

The orientation of a rigid body in three dimensions is parameterized by the Rodrigues vector $\boldsymbol{\Theta} \in \mathbb{R}^3$: its direction $\hat{\boldsymbol{\Theta}} = \boldsymbol{\Theta}/\Theta$ (with $\Theta = \|\boldsymbol{\Theta}\|$) is the rotation axis and its magnitude Θ is the rotation angle. The corresponding rotation matrix is given by the Euler–Rodrigues formula

$$\boldsymbol{\mathcal{R}}_0(\boldsymbol{\Theta}) = \cos\Theta\,\boldsymbol{1} + \sin\Theta\,[\hat{\boldsymbol{\Theta}}]_\times + (1-\cos\Theta)\,\hat{\boldsymbol{\Theta}}\otimes\hat{\boldsymbol{\Theta}}, \tag{B1}$$

with $\otimes$ the outer product. Laboratory-frame coordinates $\boldsymbol{x}$ and body-frame coordinates $\boldsymbol{X}$ are related by $\boldsymbol{x} = \boldsymbol{\mathcal{R}}_0(\boldsymbol{\theta}_0)\cdot\boldsymbol{X}$. This parameterization is singularity-free for $\Theta \in [0, 2\pi)$, unlike Euler angles, which suffer from gimbal lock.

A natural but incorrect assumption would be that the velocity $\boldsymbol{\Omega}$ equals $\dot{\boldsymbol{\Theta}}$. The correct relation follows from differentiating the Euler–Rodrigues formula with respect to time [65]

$$\boldsymbol{\Omega} = \boldsymbol{B}_\Omega(\boldsymbol{\Theta})\cdot\dot{\boldsymbol{\Theta}}, \tag{B2}$$

with the 3×3 Bortz matrix

$$\boldsymbol{B}_\Omega(\boldsymbol{\Theta}) = \frac{\Theta}{2}\,\mathrm{cotan}\!\left(\frac{\Theta}{2}\right)\boldsymbol{1} - \frac{\Theta}{2}\left[\hat{\boldsymbol{\Theta}}\right]_\times + \left(1 - \frac{\Theta}{2}\,\mathrm{cotan}\!\left(\frac{\Theta}{2}\right)\right)\hat{\boldsymbol{\Theta}}\otimes\hat{\boldsymbol{\Theta}}. \tag{B3}$$

In the limit $\Theta \to 0$, $\boldsymbol{B}_\Omega \to \boldsymbol{1}$ (using $\frac{\Theta}{2}\,\mathrm{cotan}\,\frac{\Theta}{2} \to 1$), recovering the expected identity $\boldsymbol{\Omega} \to \dot{\boldsymbol{\Theta}}$.

This relation is used to compute the six-component velocity $\boldsymbol{V}_i$ of the i-th sphere from the time-derivative of its six-component position $\boldsymbol{X}_i$ in §III B. The translational velocity of the sphere center follows directly from its position, $\boldsymbol{U}_i = \dot{\boldsymbol{R}}_i$. For the rotational part, $\boldsymbol{\Omega}_i = \boldsymbol{B}_\Omega(\boldsymbol{\Theta}_i)\cdot\dot{\boldsymbol{\Theta}}_i$. Stacking translation and rotation, the Bortz Jacobian for sphere i is the 6×6 block-diagonal matrix

$$\boldsymbol{B}_i(\boldsymbol{\Theta}_i) = \begin{bmatrix}\boldsymbol{1} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{B}_\Omega(\boldsymbol{\Theta}_i)\end{bmatrix}, \quad \text{so that} \quad \boldsymbol{V}_i = [\boldsymbol{U}_i, \boldsymbol{\Omega}_i] = \boldsymbol{B}_i(\boldsymbol{\Theta}_i)\cdot[\dot{\boldsymbol{R}}_i, \dot{\boldsymbol{\Theta}}_i]. \tag{B4}$$

The Bortz matrix $\boldsymbol{B}_\Omega$ becomes singular as $\Theta \to 2\pi$, where $\mathrm{cotan}(\Theta/2) \to \pm\infty$. To keep the integration well-conditioned, the Rodrigues vector is remapped after each time step: whenever $\|\boldsymbol{\Theta}\| \geq \pi$,

$$\boldsymbol{\Theta} \;\leftarrow\; \boldsymbol{\Theta} - 2\pi\,\hat{\boldsymbol{\Theta}}. \tag{B5}$$

This subtracts one full turn along the rotation axis. Since a rotation by Θ and a rotation by $\Theta - 2\pi$ about the same axis produce the same orientation, the physical state is unchanged. The remapping reduces the angle magnitude to $2\pi - \Theta \leq \pi$, keeping $\|\boldsymbol{\Theta}\|$ bounded away from the singularity at 2π throughout the simulation.

2. Rate-of-strain vectorization

The rate-of-strain tensor $\boldsymbol{E}_0^\infty$ is symmetric and traceless. Symmetry reduces its nine components to six independent entries, and the tracelessness condition $\mathrm{tr}(\boldsymbol{E}_0^\infty) = 0$ removes one further degree of freedom, leaving five independent components. We parameterize them as the vector

$$\boldsymbol{E} = [E_{xx},\, E_{xy},\, E_{xz},\, E_{yy},\, E_{yz}], \tag{B6}$$

from which the full tensor is recovered as

$$\boldsymbol{E}_0^\infty = \begin{pmatrix} E_{xx} & E_{xy} & E_{xz} \\ E_{xy} & E_{yy} & E_{yz} \\ E_{xz} & E_{yz} & -E_{xx} - E_{yy} \end{pmatrix}, \tag{B7}$$

with the $(3,3)$ entry $E_{zz} = -E_{xx} - E_{yy}$ enforcing tracelessness.

The double contractions $\hat{\boldsymbol{C}}_E : \boldsymbol{E}_0^\infty$ and $\boldsymbol{C}_E : \boldsymbol{E}_0^\infty$ appearing in Eqs. (17) and (32) can each be expressed as a matrix-vector product with $\boldsymbol{E}$. Because every entry $(\boldsymbol{E}_0^\infty)_{jk}$ is a linear combination of the five components E_l [Eq. (B7)], the double contraction

$$(\boldsymbol{C}_E : \boldsymbol{E}_0^\infty)_i = \sum_{j,k} (C_E)_{ijk}\, (\boldsymbol{E}_0^\infty)_{jk} \tag{B8}$$

is linear in $\boldsymbol{E}$, so it can be written as $\boldsymbol{C}_E^{(5)} \cdot \boldsymbol{E}$ for an equivalent $N_q \times 5$ matrix $\boldsymbol{C}_E^{(5)}$. The entries of $\boldsymbol{C}_E^{(5)}$ are obtained by substituting Eq. (B7) and collecting coefficients of each E_l. Because the sum runs over all nine (j,k) pairs and $\boldsymbol{E}_0^\infty$ is symmetric, each off-diagonal component (e.g. E_{xy}) contributes through both $(j,k) = (1,2)$ and $(j,k) = (2,1)$, automatically yielding the correct factor of two without requiring explicit prefactors in $\boldsymbol{E}$.

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